ADMINISTRATION SYSTÈME

OCS Inventory

Architecture 2 VMs - Installation et Configuration (Debian 12)

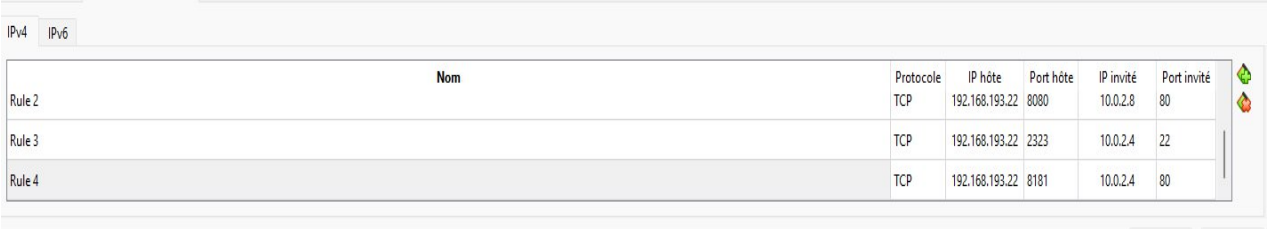
Auteur	**bthiery (Kalandy)**
VM BDD	**10.0.2.4 - MariaDB**
VM OCS	**10.0.2.8 - Apache/OCS**
Hôte Windows	**192.168.193.22**
Accès	**http://192.168.193.22:8080/ocsreports**

Partie I - Architecture et Réseau

L'installation repose sur deux machines virtuelles hébergées sur VirtualBox. Elles communiquent via un réseau NAT (NatNetwork) et sont accessibles depuis Windows via redirection de ports.

Mode Réseau	**NAT Network (NatNetwork) en 10.0.2.0/24**
IP Hôte (Windows)	**192.168.193.22**

Machine	Rôle	IP Interne	Port Interne	Port Hôte (Windows)
VM BDD	Base de données (MariaDB)	10.0.2.4	22 (SSH)	2323
VM OCS	Serveur Web (Apache/OCS)	10.0.2.8	22 (SSH)	2222
VM OCS	Interface Web	10.0.2.8	80 (HTTP)	8080



Partie II - Installation de la VM BDD (Base de Données)

Objectif : Installer MariaDB et autoriser la VM OCS à s'y connecter.

2.1 - Installation des paquets

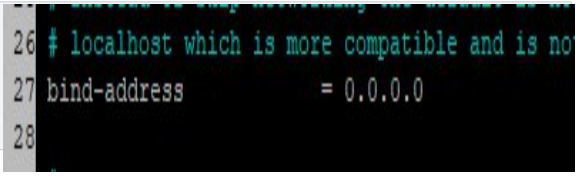
```
sudo apt update && sudo apt upgrade -y
sudo apt install -y mariadb-server
sudo mysql_secure_installation
```

2.2 - Configuration de l'écoute réseau

Par défaut, MariaDB n'écoute qu'en local. Il faut l'ouvrir au réseau.

> ➢ **Action : Éditer le fichier : sudo nano /etc/mysql/mariadb.conf.d/50-server.cnf**

```
# Remplacer :
bind-address = 127.0.0.1
# Par :
bind-address = 0.0.0.0
```




```
sudo systemctl restart mariadb
```

2.3 - Création de la base et de l'utilisateur

Créer un utilisateur qui a le droit de se connecter depuis l'IP de la VM OCS (10.0.2.8).

```
sudo mysql -u root -p

CREATE DATABASE ocsdatabase;
CREATE USER 'ocs-user'@'10.0.2.8' IDENTIFIED BY 'ocs-mdp';
GRANT ALL PRIVILEGES ON ocsdatabase.* TO 'ocs-user'@'10.0.2.8';
FLUSH PRIVILEGES;
EXIT;
```

Partie III - Installation de la VM OCS (Serveur d'Application)

Objectif : Installer le moteur OCS et le connecter à la BDD distante.

3.1 - Installation des dépendances

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y apache2 libapache2-mod-perl2 make gcc \
  php php-mysql php-gd php-xml php-mbstring php-curl libxml-simple-
perl \
  libdbi-perl libdbd-mysql-perl libapache-dbi-perl libnet-ip-perl \
  libarchive-zip-perl libmojolicious-perl libswitch-perl libplack-perl
\
  mariadb-client
```

3.2 - Installation OCS Inventory

```
wget
https://github.com/PassAndSecure/OCS_inventory/releases/download/OCSNG_UNIX_SERVER
-2.12.1/OCSNG_UNIX_SERVER-2.12.1.tar.gz
tar -xvzf OCSNG_UNIX_SERVER-2.12.1.tar.gz
cd OCSNG_UNIX_SERVER-2.12.1
sudo ./setup.sh
```

> ⚠ **Attention :** *Durant l'installation, à la question "Database Server Host", entrer l'IP interne de la BDD : 10.0.2.4.*

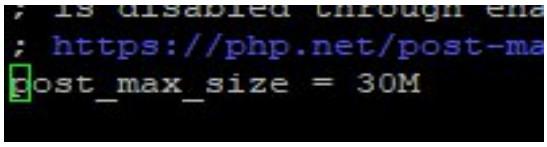
3.3 - Configuration Apache et Permissions

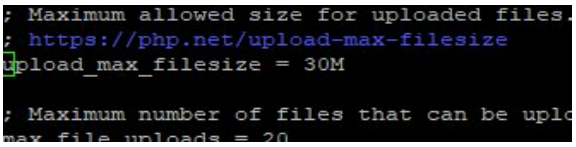
```
sudo a2enconf ocsinventory-reports
sudo chown -R www-data:www-data /usr/share/ocsinventory-reports
sudo chmod -R 755 /usr/share/ocsinventory-reports
sudo chown -R www-data:www-data /var/lib/ocsinventory-reports
sudo chmod -R 755 /var/lib/ocsinventory-reports
```

3.4 - Configuration PHP (Taille d'upload)

➤ Action : Éditer : sudo nano /etc/php/8.2/apache2/php.ini

```
post_max_size = 30M
upload_max_filesize = 30M
```





3.5 - Configuration de la connexion BDD (Serveur de communication)

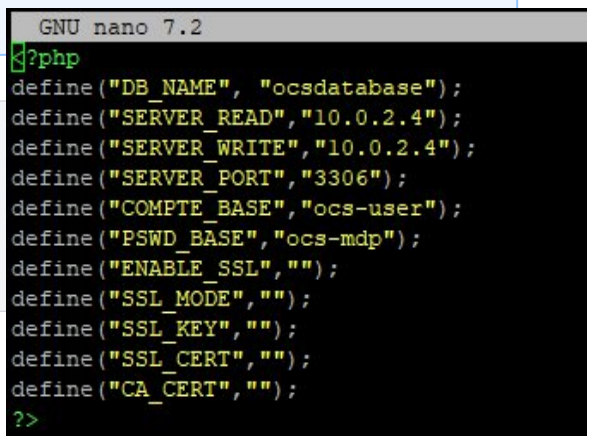
➤ Action : Éditer : sudo nano /etc/apache2/conf-available/z-ocsinventory-server.conf

```
PerlSetEnv OCS_DB_HOST 10.0.2.4
PerlSetEnv OCS_DB_PORT 3306
PerlSetEnv OCS_DB_NAME ocsdatabase
PerlSetEnv OCS_DB_LOCAL ocsdatabase
PerlSetEnv OCS_DB_USER ocs-user
PerlSetVar OCS_DB_PWD ocs-mdp
```

3.6 - Configuration de la connexion BDD (Interface Web)

➤ Action : Éditer : sudo nano /usr/share/ocsinventory-reports/ocsreports/dbconfig.inc.php

```
define("DB_NAME", "ocsdatabase");
define("SERVER_READ","10.0.2.4");
define("SERVER_WRITE","10.0.2.4");
define("SERVER_PORT","3306");
define("COMPTE_BASE","ocs-user");
define("PSWD_BASE","ocs-mdp");
```



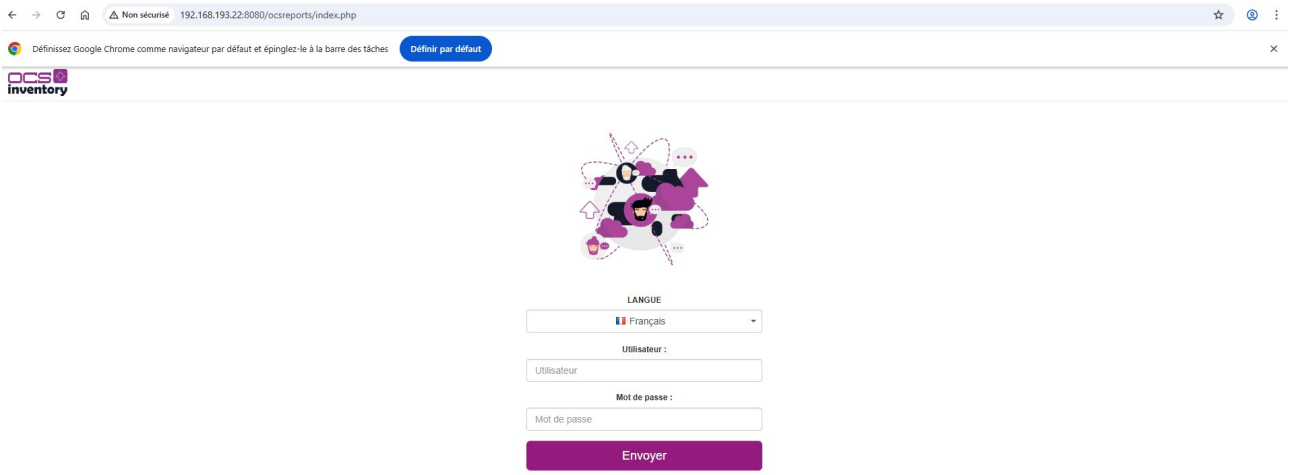
3.7 - Redémarrage final

```
sudo systemctl restart apache2
```

Partie IV - Finalisation et Accès

4.1 - Accès Web

➤ **Action : Depuis le poste Windows, ouvrir : http://192.168.193.22:8080/ocsreports**

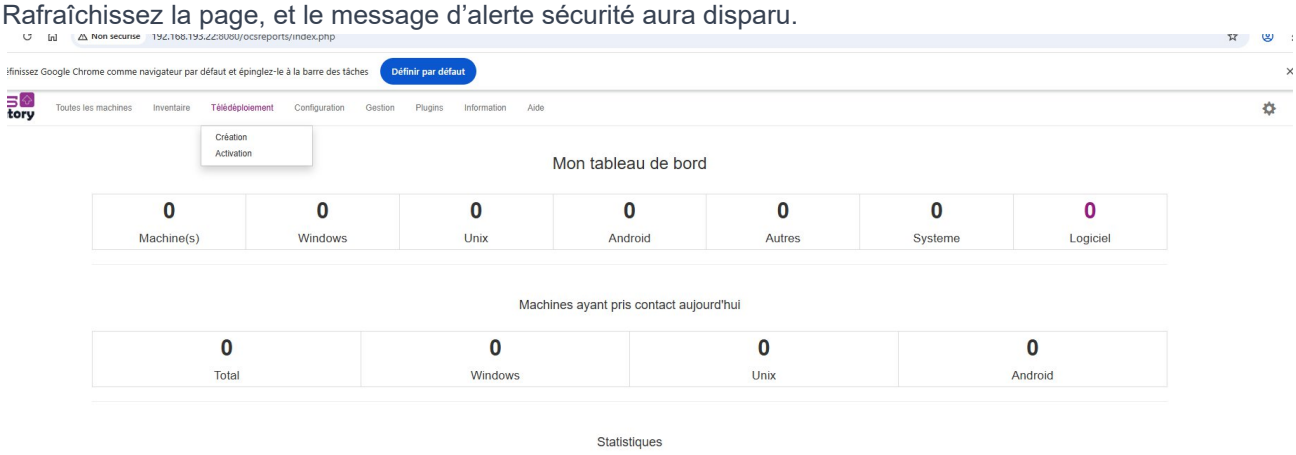


4.2 - Assistant de configuration

MySQL Host	**10.0.2.4**
MySQL User	**ocs-user**
MySQL Password	**ocs-mdp**
Database Name	**ocsdatabase**

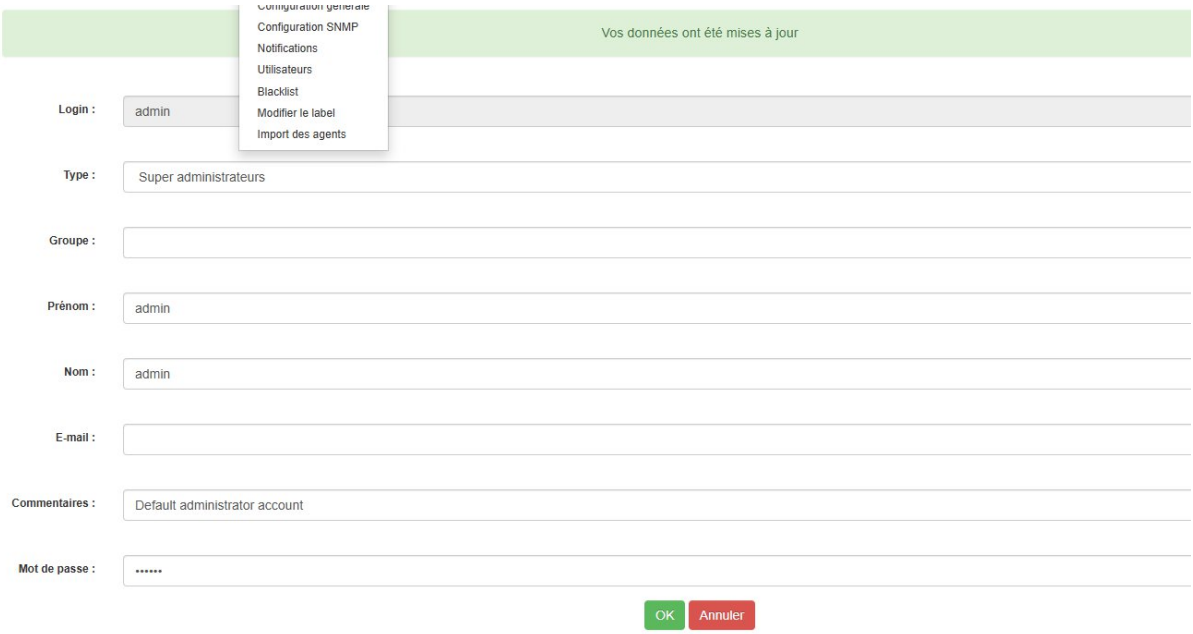
Une fois la mise à jour terminée :

1. Supprimer le fichier d'installation : sudo rm /usr/share/ocsinventory-reports/ocsreports/install.php
2. Rafraîchissez la page, et le message d'alerte sécurité aura disparu.



3. Se connecter avec admin / admin

4. Changer le mot de passe immédiatement



Partie V - Mise en Place du HTTPS (Serveur Web)

5.1 - Activation du module SSL

```
sudo a2enmod ssl
sudo systemctl restart apache2
```

5.2 - Génération d'un certificat auto-signé

```
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
-keyout /etc/ssl/private/apache-selfsigned.key \
-out /etc/ssl/certs/apache-selfsigned.crt
```

5.3 - Configuration du site SSL Apache

➤ **Action : Éditer : sudo nano /etc/apache2/sites-available/default-ssl.conf**

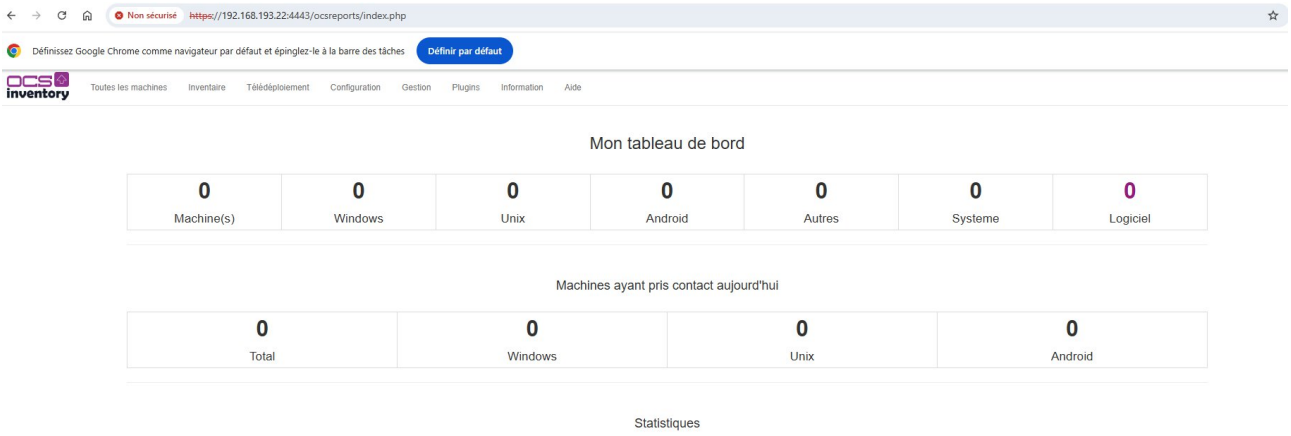
```
31        SSLCertificateFile      /etc/ssl/certs/apache-selfsigned.crt
32        SSLCertificateKeyFile   /etc/ssl/private/apache-selfsigned.key
```

```
SSLCertificateFile /etc/ssl/certs/apache-selfsigned.crt
SSLCertificateKeyFile /etc/ssl/private/apache-selfsigned.key
```

```
sudo a2ensite default-ssl
sudo systemctl reload apache2
```

5.4 - Configuration VirtualBox pour HTTPS (port 4443)

IP Hôte	**192.168.193.22**
Port Hôte	**4443**
IP VM OCS	**10.0.2.8**
Port Invité	**443**



✓ **Résultat :** *Accès HTTPS : https://192.168.193.22:4443/ocsreports*

Partie VI - Sécurisation de la Liaison BDD (SSL/TLS)

Objectif : Chiffrer le flux de données entre la VM OCS et la VM BDD.

6.1 - Génération des clés sur la VM BDD

```
sudo mkdir -p /etc/mysql/ssl
cd /etc/mysql/ssl
# CA :
sudo openssl genrsa 2048 > ca-key.pem
sudo openssl req -new -x509 -nodes -days 3650 -key ca-key.pem -out ca-
cert.pem -subj "/CN=MariaDB-CA"
# Certificat Serveur (CN=10.0.2.4) :
sudo openssl req -newkey rsa:2048 -days 3650 -nodes -keyout server-
key.pem -out server-req.pem -subj "/CN=10.0.2.4"
sudo openssl rsa -in server-key.pem -out server-key.pem
sudo openssl x509 -req -in server-req.pem -days 3650 -CA ca-cert.pem -
CAkey ca-key.pem -set_serial 01 -out server-cert.pem
# Permissions :
sudo chown -R mysql:mysql /etc/mysql/ssl
sudo chmod 600 /etc/mysql/ssl/*.pem
```

6.2 - Configurer MariaDB pour forcer SSL

➤ Action : Éditer : sudo nano /etc/mysql/mariadb.conf.d/50-server.cnf - section [mysqld]

```
ssl-ca = /etc/mysql/ssl/ca-cert.pem
ssl-cert = /etc/mysql/ssl/server-cert.pem
ssl-key = /etc/mysql/ssl/server-key.pem
require_secure_transport = ON
```

```
GNU nano 7.2                    /etc/mysql/mariadb.conf.d/50-server.cnf *
#log_slow_min_examined_row_limit = 1000

# The following can be used as easy to replay backup logs or for replication.
# note: if you are setting up a replication slave, see README.Debian about
#       other settings you may need to change.
#server-id              = 1
#log_bin                = /var/log/mysql/mysql-bin.log
expire_logs_days        = 10
#max_binlog_size        = 100M

#
# * SSL/TLS
#

# For documentation, please read
# https://mariadb.com/kb/en/securing-connections-for-client-and-server/
ssl-ca = /etc/mysql/ssl/ca-cert.pem
ssl-cert = /etc/mysql/ssl/server-cert.pem
ssl-key = /etc/mysql/ssl/server-key.pem
require-secure-transport = ON
```

```
sudo systemctl restart mariadb
```

Partie VII - Configuration Client OCS pour SSL

7.1 - Copier le certificat CA sur la VM OCS

```
sudo mkdir -p /etc/ocsinventory-reports/ssl
sudo nano /etc/ocsinventory-reports/ssl/ca-cert.pem  # coller le
contenu de ca-cert.pem
sudo chown -R www-data:www-data /etc/ocsinventory-reports/ssl
sudo chmod 600 /etc/ocsinventory-reports/ssl/ca-cert.pem
```

```
  GNU nano 7.2                                  /etc/ocsinventory-reports/ssl/ca-cert.pem
-----BEGIN CERTIFICATE-----
MIIDCzCCAfOgAwIBAgIUEyPKGepfBVfPjoYz4CqzZ51GGcgwDQYJKoZIhvcNAQEL
BQAwFTETMBEGA1UEAwwKTWFyaWFEQi1DQTAeFw0yNTEyMDgxMDA2MDBaFw0zNTEy
MDYxMDA2MDBaMBUxEzARBgNVBAMMCk1hcmlhREItQ0EwggEiMA0GCSqGSIb3DQEB
AQUAA4IBDwAwggEKAoIBAQDk8Hv9h2R0s06gYDSIab0BxN3vUJmMu2RYY5Av1xl5
tGLDB2KnG2uylfPgrr37oBtRhkvnfDIv4JYLRLSn3Bu+YuSZD4qxcdRaQfgF4/QK
EftLe1w0TvgO+ciKb1T7pmGXCsnNAiIxZYjZUDgAcvdCZxVKdXkU7buYCxkqw1WA
exh6QPzsH3BMi2bfn0JfuDwR4N8Rp2PLQ8+6y+Pja6rbCars7j1Tj45c0qKwodWk
C37X9ybqoYqSp2tqztVFIA6c7pf8h8R1F1SsfwHzKcp6qKF72mVloqr0hiWuLRfJ
BsvJ34udC1MBm5pmpny3vVXjKtaPqgn8ysDmI9h08dFFAgMBAAGjUzBRMB0GA1Ud
DgQWBBTuHM0ZM1L7iFcB4k42OmrA8qaq8jAfBgNVHSMEGDAWgBTuHM0ZM1L7iFcB
4k42OmrA8qaq8jAPBgNVHRMBAf8EBTADAQH/MA0GCSqGSIb3DQEBCwUAA4IBAQDA
zUHh7204oNyPAh3Gs6NODWV1kocJ15CVCDyA4hIyPmemPFz88pM6D6PHb4CEO/+/
6EjoFvxXd+gwvBa+1IIU7ebXxAII/mw0PHD5FSRN54vwt0nr/iP2Mi49uzuEJCe4
lLGyAb8ScYZ6XbFr9eKQttLqGYSHf5BGFrbBQ3sFUcDeg/Us2/JSNdPaqPiwrVze
MJkUqz/UaSpPBlvlgspkh2k8oPvuOb5oHOlP90fe/LVxw2aCmlL0bgNlGfX9nwqT
xQ8hY/ncnMIyUVtvVTsaEi/lFwPl2RHq5QYPQqveZR6bpS97nvzhaOFl308rWnly
QthX5GkRo6Dm6e5J9MK/
-----END CERTIFICATE-----
```

7.2 - Activer SSL dans dbconfig.inc.php

```
define("ENABLE_SSL","1");
define("SSL_MODE","SSL_MODE_REQUIRED");
define("CA_CERT","/etc/ocsinventory-reports/ssl/ca-cert.pem");
```

```
  GNU nano 7.2                    /usr/share/ocsinventory-reports/ocsreports/dbconfig.inc.php *
<?php
define("DB_NAME", "ocsdatabase");
define("SERVER_READ","10.0.2.4");
define("SERVER_WRITE","10.0.2.4");
define("SERVER_PORT","3306");
define("COMPTE_BASE","ocs-user");
define("PSWD_BASE","ocs-mdp");
define("ENABLE_SSL","");
define("SSL_MODE","SSL_MODE_REQUIRED");
define("SSL_KEY","");
define("SSL_CERT","");
define("CA_CERT","/etc/ocsinventory-reports/ssl/ca-cert.pem");
?>
```

7.3 - Configuration du Serveur de Communication Perl

```
PerlSetEnv OCS_DB_SSL_ENABLED 1
PerlSetEnv OCS_DB_SSL_CA /etc/ocsinventory-reports/ssl/ca-cert.pem
PerlSetEnv OCS_DB_SSL_MODE SSL_MODE_REQUIRED
```

```
35  PerlSetEnv OCS_DB_SSL_ENABLED 1
36  # PerlSetEnv OCS_DB_SSL_CLIENT_KEY /etc/ssl/private/client.key
37  # PerlSetEnv OCS_DB_SSL_CLIENT_CERT /etc/ssl/certs/client.crt
38  PerlSetEnv OCS_DB_SSL_CA_CERT /etc/ocsinventory-reports/ssl/ca-cert.pem
39  # SSL Mode
40  # - SSL_MODE_PREFERRED (SSL enabled but optional)
41  # - SSL_MODE_REQUIRED (SSL enabled, mandatory but don't verify server certificate. Ex self signed cert)
42  # - SSL_MODE_STRICT (SSL enabled, mandatory and server cert must be trusted)
43  PerlSetEnv OCS_DB_SSL_MODE SSL_MODE_REQUIRED
```

```
sudo systemctl restart apache2
```

Partie VIII - Vérification et Preuves de Fonctionnement

Pour valider que le chiffrement est actif entre les deux VM :

```
sudo mysql -u root -p
SHOW STATUS LIKE 'Ssl_cipher';
SHOW GLOBAL VARIABLES LIKE 'require_secure_transport';
```

```
MariaDB [(none)]> SHOW STATUS LIKE 'Ssl_cipher'; SHOW GLOBAL VARIABLES LIKE 'require_secure_transport';
+---------------+-----------------------+
| Variable_name | Value                 |
+---------------+-----------------------+
| Ssl_cipher    | TLS_AES_256_GCM_SHA384 |
+---------------+-----------------------+
1 row in set (0,001 sec)

+--------------------------+-------+
| Variable_name            | Value |
+--------------------------+-------+
| require_secure_transport | ON    |
+--------------------------+-------+
1 row in set (0,000 sec)
```

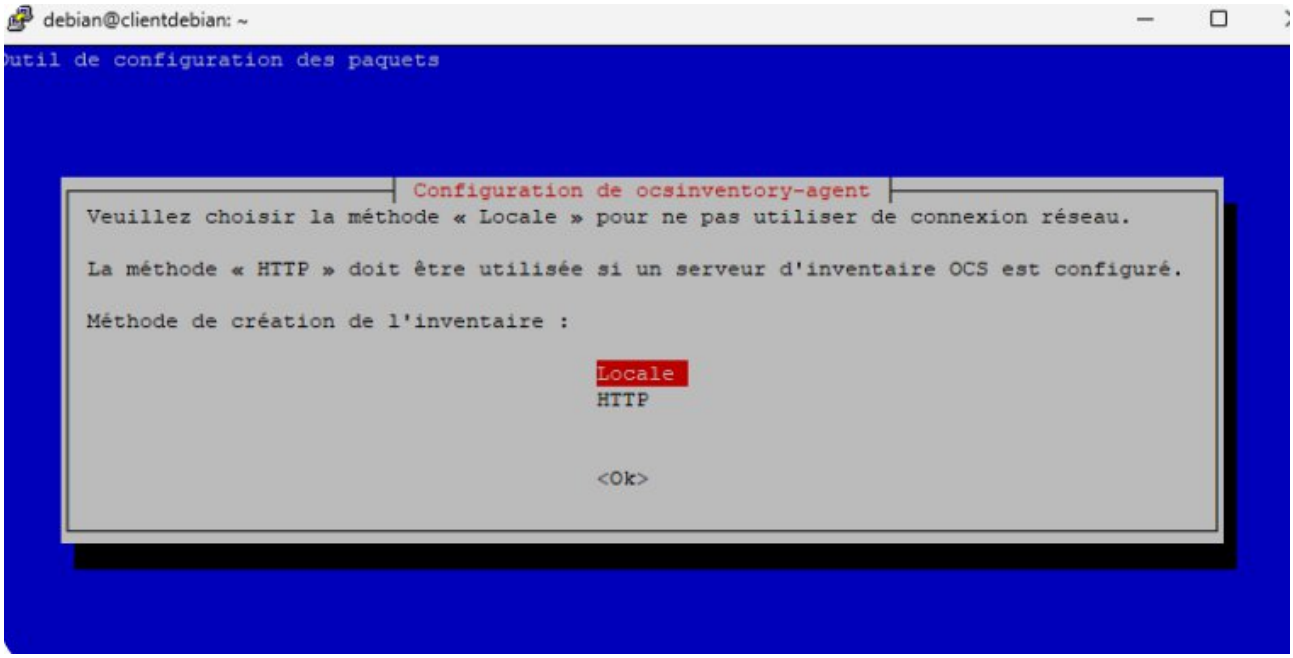
✓ **Résultat :** *Le chiffrement SSL est actif entre la VM OCS et la VM BDD.*

Partie IX - Installation du Client Debian

9.1 - Installation de l'agent

```
sudo apt update
sudo apt install ocsinventory-agent -y

(Si l'assistant de configuration apparaît, choisissez "HTTP" et
laissez les valeurs par défaut, nous allons les corriger via le
fichier).
```



9.2 - Configuration de l'agent

➢ **Action : Éditer : sudo nano /etc/ocsinventory/ocsinventory-agent.cfg**

```
server=https://10.0.2.8/ocsinventory
ssl=1
ca=/etc/ocsinventory/server.crt
basevardir=/var/lib/ocsinventory-agent
debug=1
no_proxy=10.0.2.8
```

9.3 - Test de l'inventaire

```
sudo ocsinventory-agent --debug
```

✓ **Résultat :** *Succès : lignes "[debug] sending XML" puis "[debug] sent: 200 OK" visibles.*

Partie X - Installation du Client Windows

10.1 - Préparation du certificat

Récupérer le texte du certificat depuis la VM OCS, le coller dans le Bloc-notes Windows et l'enregistrer sous server.crt.

10.2 - Configuration de l'agent Windows

OCS Server URL	**https://10.0.2.8/ocsinventory**
Validate Certificates	**COCHER cette case (Obligatoire)**
CA Certificate File	**C:\server.crt**
Proxy Type	**None**

Partie XI - Installation de GLPI

11.1 - Dépendances sur la VM GLPI (10.0.2.9)

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y apache2 php php-curl php-gd php-intl php-mbstring
php-soap php-xml php-zip php-bz2 php-mysql libapache2-mod-php wget
```

11.2 - Créer la base GLPI (VM BDD)

```
sudo mysql -u root -p
CREATE DATABASE glpidb;
GRANT ALL PRIVILEGES ON glpidb.* TO 'ocs-user'@'10.0.2.%';
FLUSH PRIVILEGES;
EXIT;
```

11.3 - Installer GLPI

```
cd /tmp
wget https://github.com/glpi-
project/glpi/releases/download/10.0.10/glpi-10.0.10.tgz
tar -xzvf glpi-10.0.10.tgz
sudo mv glpi /var/www/html/
sudo chown -R www-data:www-data /var/www/html/glpi
```

Partie XII - Liaison OCS - GLPI

Faire apparaître les machines remontées par OCS dans l'interface de gestion GLPI.

1. Dans GLPI, menu Configuration > Plugins > Marketplace > Installer "OCS Inventory NG"
2. Menu Outils > OCS Inventory NG > Ajouter un serveur OCS
3. Hôte BDD : 10.0.2.4 / Base : ocsdatabase / Utilisateur : ocs-user
4. Tester la connexion puis importer les machines

Mission OCS Inventory complétée - Architecture 2 VMs + GLPI + SSL opérationnels.

- Fin de la documentation -